

- 2 The Galápagos Islands are a group of islands in the Pacific Ocean that are approximately 900 km from Ecuador in South America. Charles Darwin developed his theory of evolution by natural selection after visiting the Galápagos Islands in the 1830s.

Finches are small birds that are commonly found on the Galápagos Islands.

Some biologists studied one species of finch, *Geospiza conirostris*, living on Genovesa Island in the Galápagos. Genovesa Island is a small island with no mountains.

Figure 2.1 shows *G. conirostris*.



Figure 2.1

The biologists studied several breeding pairs of *G. conirostris*. Adult male *G. conirostris* sing a song to attract females for mating. Female *G. conirostris* do **not** sing.

The biologists noticed that adult male *G. conirostris* could be separated into two groups based on their song type. Adult male *G. conirostris* sing either song type **A** or song type **B**.

At the start of the investigation, the biologists caught the birds in a mist net.

Figure 2.2 shows a bird caught in a mist net. The mist net does not harm the bird.



Figure 2.2

For each male *G. conirostris* caught in the mist net, the biologists:

- attached a coloured ring to the leg of the bird so that it could be easily identified in the future
- measured the mass of the bird
- measured the depth, width and length of the bill of the bird
- released the bird back into the ecosystem
- observed the song type and other behaviours of the bird over a period of four months.

(a) Table 2.1 shows the data from the investigation.

Table 2.1

song type of male <i>G. conirostris</i>	number of birds	mean mass /g	mean bill depth /mm	mean bill width /mm	mean bill length /mm
song type A	11	25.85	10.79	9.99	15.47
song type B	13	24.96	11.02	10.23	14.55

Statistical analysis showed that there was no significant difference between the mean mass, the mean bill depth and mean bill width of male *G. conirostris* with song type **A** and male *G. conirostris* with song type **B**.

The biologists then carried out a *t*-test to compare the mean bill lengths of male *G. conirostris* with song type **A** and male *G. conirostris* with song type **B**.

(i) State **two** conditions of the data that allowed the biologists to carry out a *t*-test.

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(ii) State the null hypothesis for the *t*-test that the biologists carried out.

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(iii) Calculate the number of degrees of freedom (ν) for the *t*-test that the biologists carried out.

number of degrees of freedom = [1]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

(iv) The biologists calculated the value of t as **3.768**.

Table 2.2 shows the probability table for the t -test.

Table 2.2

degrees of freedom (ν)	critical values		
	$p = 0.50$	$p = 0.10$	$p = 0.05$
20	0.687	1.725	2.086
21	0.686	1.721	2.080
22	0.686	1.717	2.074
23	0.685	1.714	2.069
24	0.685	1.711	2.064
25	0.684	1.708	2.060
26	0.684	1.706	2.056

Use the calculated value of t **and** Table 2.2 to explain what can be concluded about the mean bill lengths of male *G. conirostris* with song type **A** and male *G. conirostris* with song type **B**.

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(b) *G. conirostris* feed on a range of food sources, including:

- the leaves, flowers and fruits of the prickly pear cactus
- the seeds of the palo santo tree.

Initial observations suggested that male *G. conirostris* with song type **A** feed on different food sources than male *G. conirostris* with song type **B**.

Suggest a method that the biologists could use to measure the proportion of time spent feeding on each food source by male *G. conirostris* with song type **A** and male *G. conirostris* with song type **B**.

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(c) The biologists concluded that *G. conirostris* on Genovesa Island may be in the early stages of sympatric speciation.

- supports the conclusion that *G. conirostris* may be in the early stages of sympatric speciation
- could be extended to increase confidence in the conclusion.

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[4]

[Total: 13]

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